

Question	Answer	Marks
2(a)	gas that obeys equation $pV = \text{constant} \times T$	M1
	symbols p, V and T explained	A1
2(b)(i)	$pV = \frac{1}{3} Nm\langle c^2 \rangle$ and $M = Nm$ (and so) $p = \frac{1}{3}\rho\langle c^2 \rangle$	C1
	$2.12 \times 10^7 = \frac{1}{3} \times [3.20 / (1.84 \times 10^{-2})] \times \langle c^2 \rangle$	C1
	$c_{\text{r.m.s.}} = 605 \text{ m s}^{-1}$	A1
2(b)(ii)	1. $pV = nRT$ and $T = (22 + 273)\text{K}$	C1
	$n = (2.12 \times 10^7 \times 1.84 \times 10^{-2}) / (8.31 \times 295)$ $= 159 \text{ mol}$	A1
	2. mass = $3.20 / (159 \times 6.02 \times 10^{23})$ or mass = $[2 \times (3/2) \times 1.38 \times 10^{-23} \times 295] / 605^2$	C1
	mass = $3.34 \times 10^{-26} \text{ kg}$	A1
2(c)	$A = (3.34 \times 10^{-26}) / (1.66 \times 10^{-27})$ $= 20$	A1

9702/42

Cambridge International AS/A Level – Mark Scheme

October/November 2018

PUBLISHED

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2(c)	(thermal) energy per unit mass (to cause change of state)	D1